

```
$ spring run TransactionLab --propagation=REQUIRED
```

@Transactional



// 数据库事务 — 数据库事务

begin

开始

commit

提交 flush

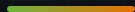
rollback

回滚

proxy

代理

\$ cat agenda.md



01 /



□□□□□□□□

02 /



□□□□□□□□

03 / AOP □□

□□□□□□□□

04 /



□□□□rollback □□□FAQ

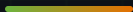
🔗🔗🔗🔗🔗🔗🔗 @Transactional □□□□

PART 01

@Transactional □□□□

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□□□□□□□□



1. 

□□□□

2. A□□

UPDATE accounts...

3. B□□

UPDATE accounts...

4. 

Commit

WHAT IF STEP 3 FAILS?

□□□□□□ — □□□□□□□□□□Rollback□□□□□ A□

@Transactional □□□□□□□□□□□□□□□□□□□□□□□□

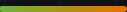
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□□□□□□□□□□ → commit → □□ flush() □ □□□□□□□□ Entity □□□ save()□□□□□□□□□

PART 02



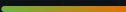


```
1 // 增删改查
2 // 增删改查
3 add*, save*, create*,
4 update*, delete*
```

PROPAGATION_REQUIRED

增删改查增删改查增删改查增删改查

☐ RuntimeException → ☐☐

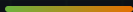


```
1 // 增删改查
2 // 增删改查
3 find*, get*, search*,
4 getCount*, *增删改查
```

PROPAGATION_NOT_SUPPORTED

增删改查增删改查增删改查增删改查

setReadOnly(true)☐☐☐☐



```
1 // ✅ 测试
2 public void saveUser() { ... }
3 public void updateOrder() { ... }
4
5 // ❌ 测试@Transactional
6 public void processUser() { ... } // 测试
7 public void handleOrder() { ... } // * 测试
```

PRIORITY

测试 @Transactional 测试 process* 测试

测试测试测试测试

CHECKPOINT_01

□□□□□□□□□□□□□□□□

A processUser() @Transactional□□□□□□□□

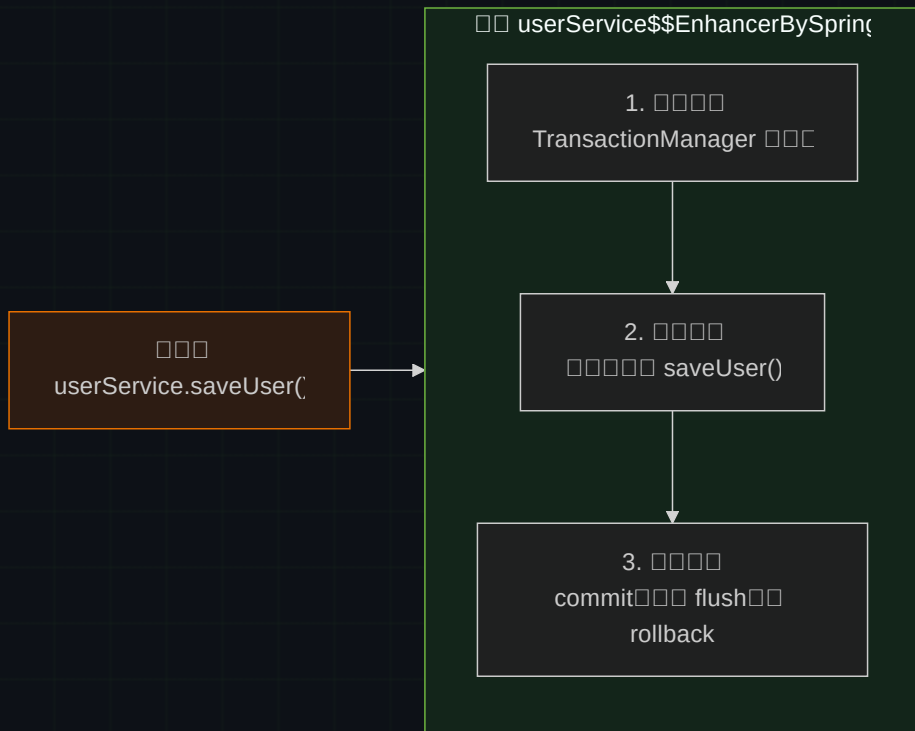
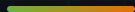
B □□□□ this.saveUser() □□□□□□□ AOP □□

C □□□□□□□□□□□□□□□□process* □□□□□□□□

PART 03

AOP □□□□□□□

Spring □□□□□□□□□□



PART 04



90% ██████████



```
1  @Service
2  public class UserService {
3
4      // 🚀 快速实现
5      public void processAndSave(User user) {
6          validateUser(user);
7          saveUser(user); // this.saveUser()
8      }
9
10     public void saveUser(User user) {
11         userRepository.save(user);
12     }
13 }
```

WHY

`this.saveUser()` 需要调用 `saveUser()` 方法 --
AOP 需要调用

1. 需要调用 `Service`

2. 需要调用 `self.saveUser()`

```
□□□□□□□process* □□□
```

this.

```
private 000000000000
```

NDGHHH
NDGHHH
NDGHHH
NDGHHH
NDGHHH

\$ ████████████████████████████████ 90%



DO

- ✓ 使用 save* / update* / delete*
- ✓ 使用事务
- ✓ 使用 Service 接口

DON'T

- × 使用 DAO
- × 使用 Controller
- × 使用 Entity



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□□□□□commit NOT DESIGN NOT DESIGN NOT DESIGN flush NOT DESIGN NOT DESIGN rollback

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□□□□□□

this. NOT DESIGN NOT DESIGN NOT DESIGN private □□□□□

□□□□

NOT DESIGN NOT DESIGN NOT DESIGN find* / get* NOT DESIGN NOT DESIGN NOT DESIGN dirty checking

□□□□□□□□ — □□□□□□□□

Transaction committed.

[[[JPA flush dirty checking]]

```
$ spring run TransactionLab --status
```

```
begin → invoke → commit(flush) | rollback
```